

On the decomposition of a strong epimorphism into regular epimorphisms

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1 The decomposition number

2 Minimum vs. canonical

Regular-epi decomposition

Recall: the fundamental theorem of homomorphisms

For $A \xrightarrow{f} B$,

$$A / \text{Ker}(f) \xrightarrow[\text{the coeq. of the kernel pair}]{A / \text{Ker}(f)} \cong \text{Im}(f)$$

the smallest s

Equivalently, the unique morphism f_1 becomes mono:

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \searrow \text{reg. epi} & & \nearrow f_1 \\ & \text{Coim } f := A / \text{Ker } f & \end{array}$$

The theorem holds in any category of equational algebras.

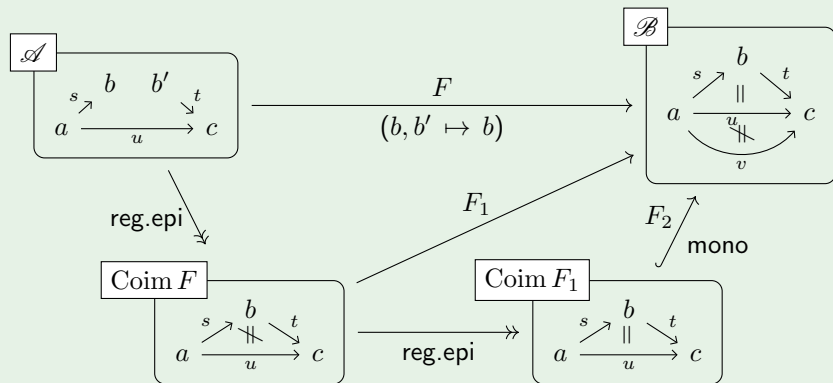
Question

Does the theorem hold in any category? $\rightsquigarrow$ No.

The case of Cat

Counterexample

In Cat,



The decomposition number

Definition [GU71; MS82; Bör91; KN26]

- 1 The (canonical) decomposition number $\sigma(f) :=$ the smallest α s.t. " f_α " becomes mono.
- 2 For a category $\mathcal{C}$, $\sigma(\mathcal{C}) :=$ the smallest β s.t. $\forall f$ in $\mathcal{C}$, $\sigma(f) < \beta$.

Example

- 1 $\sigma(\mathbf{Set}) = \sigma(\mathbf{Grp}) = 2$. (the fundamental thm. of homomorphisms)
- 2 $\sigma(\mathbf{Pos}) = 2$.
- 3 $\sigma(\mathbf{Field}) = 1$.
- 4 $\sigma(\mathbf{Cat}) = 3$ [BBP99].

Theorem [GU71]

$\mathcal{C}$: locally κ -presentable $\rightsquigarrow \sigma(\mathcal{C}) \leq \kappa + 1$.

However, this only provides a lough upper bound $\sigma(\mathbf{Cat}) \leq \omega + 1$.

Giving a tight upper bound

$\mathcal{C}$: locally presentable

Task: Find a stricter upper bound for $\sigma(\mathcal{C})$.

Strategy:

- 1 Construct a logical theory $\mathbb{T}$ s.t. $\mathbf{Mod} \mathbb{T} \simeq \mathcal{C}$.
- 2 Looking closely at $\mathbb{T}$, we would obtain an upper bound.

Theorem (informal) [KN26]

This strategy works well if $\mathbb{T}$ is a *partial Horn theory* [PV07] or a *generalized algebraic theory* [Car86].

Application

$$\begin{aligned}\sigma(n\text{-Cat}) &\leq n + 2, & \sigma(\mathbf{StrMonCat}) &\leq 3 \\ \sigma(\mathbf{MultiCat}) &\leq 3, & \sigma(\mathbf{DblCat}) &\leq 4.\end{aligned}$$

In fact, these are equalities!

1 The decomposition number

2 Minimum vs. canonical

Minimum vs. canonical

Recall

$\sigma(f) :=$ the “length” of the **canonical** decomp. of f . (taking the coimage)

How does this relate to **arbitrary** decompositions?

Definition [KN26]

The **minimum decomposition number** $\delta(f) :=$ the shortest possible length of decompositions of f .

Theorem [KN26]

In a category with kernel pairs and colimits,

$$\sigma(f) = \delta(f).$$

That is, “**canonical = minimum.**” (This can be verified easily.)

Further generalization

We started from:

- regular epimorphisms, and
- the canonical decompositions under them.

→ These can be generalized to a class $\mathbf{E}$ of morphisms that admits *locally orthogonal $\mathbf{E}$ -factorizations* [Tho79; MT82; Tho83].

Example

- 1 $\mathbf{E} := \{\text{strict localizations in } \mathbf{Cat}\}.$
- 2 $\mathbf{E} := \{\text{monotone quotient maps in } \mathbf{Top}\}.$
- 3 $\mathbf{E} := \{\text{strictly monadic functors in } \mathbf{Cat}_{\text{radj}}\}.$

Theorem [KN26]

For a class $\mathbf{E}$ of morphisms that admits locally orthogonal $\mathbf{E}$ -factorizations,

- 1 If $\mathbf{E} \subseteq \{\text{epis}\}$, then $\sigma(f) = \delta(f)$.
- 2 If $\mathbf{E} \not\subseteq \{\text{epis}\}$, then $\sigma(f) = \delta(f)$ but only for finite ordinals. (Non-trivial)

Thank you!

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